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(54) **COMPOUND CURVE CABLE CHAIN**

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CPC ***A61B 6/4441*** (2013.01); ***A61B 6/03*** (2013.01); ***A61B 6/4405*** (2013.01)

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CPC A61B 6/035; A61B 6/4405; A61B 6/4441
See application file for complete search history.

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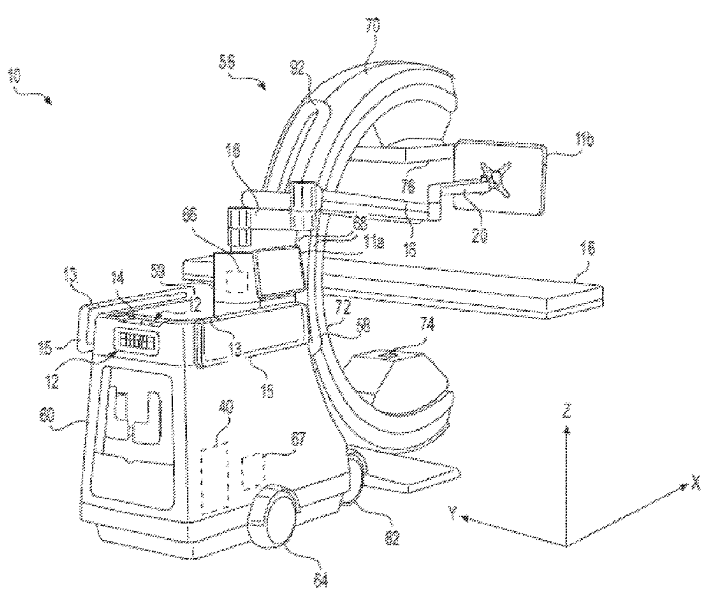
(57) **ABSTRACT**

Embodiments generally relate to routing a bundle of loose cables with a cable chain during a medical procedure. The cable chain comprises split tubing and a plurality of links extending discretely along a length of the split tubing. Each link comprises a housing including an outer surface and an inner surface. The outer surface comprises a magnet and the inner surface forms a recess in the link. The split tubing is disposed within the recesses of the links.

20 Claims, 18 Drawing Sheets



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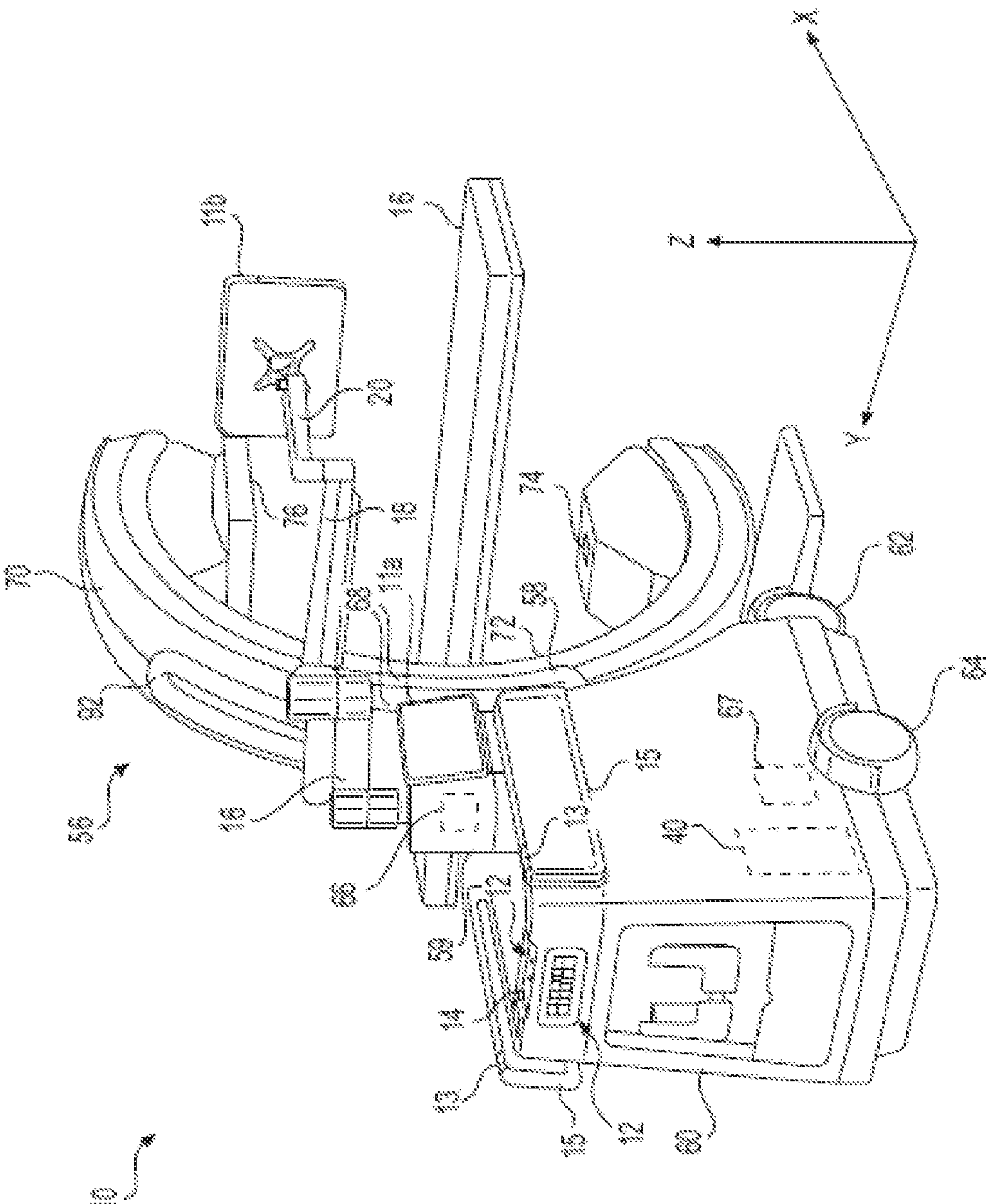
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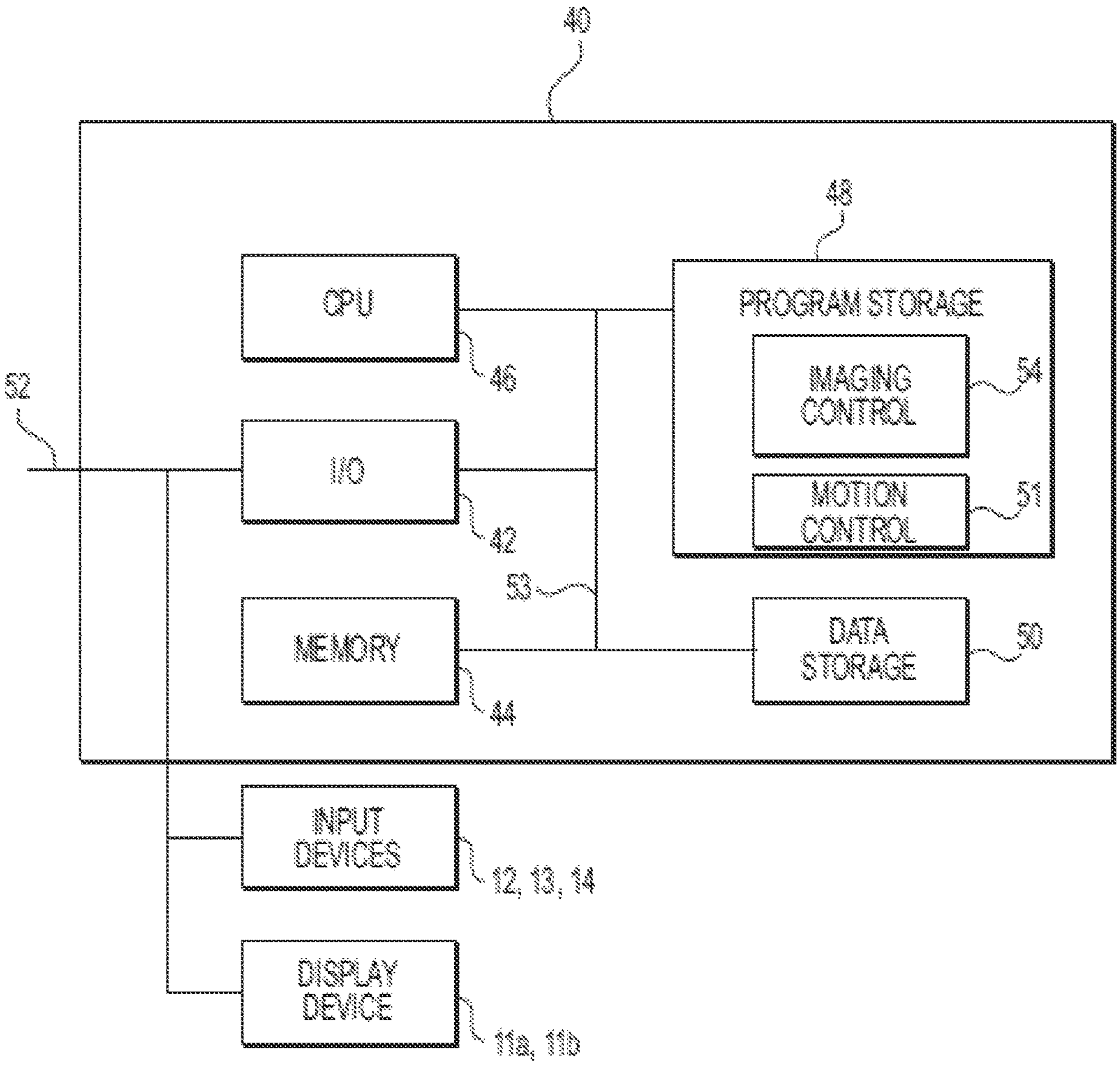
FIG. 1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IMAGING CONTROL
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52
I/O
42
MOTION CONTROL
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MEMORY
44
DATA STORAGE
50
INPUT DEVICES
12, 13, 14
DISPLAY DEVICE
11a, 11b

FIG. 2

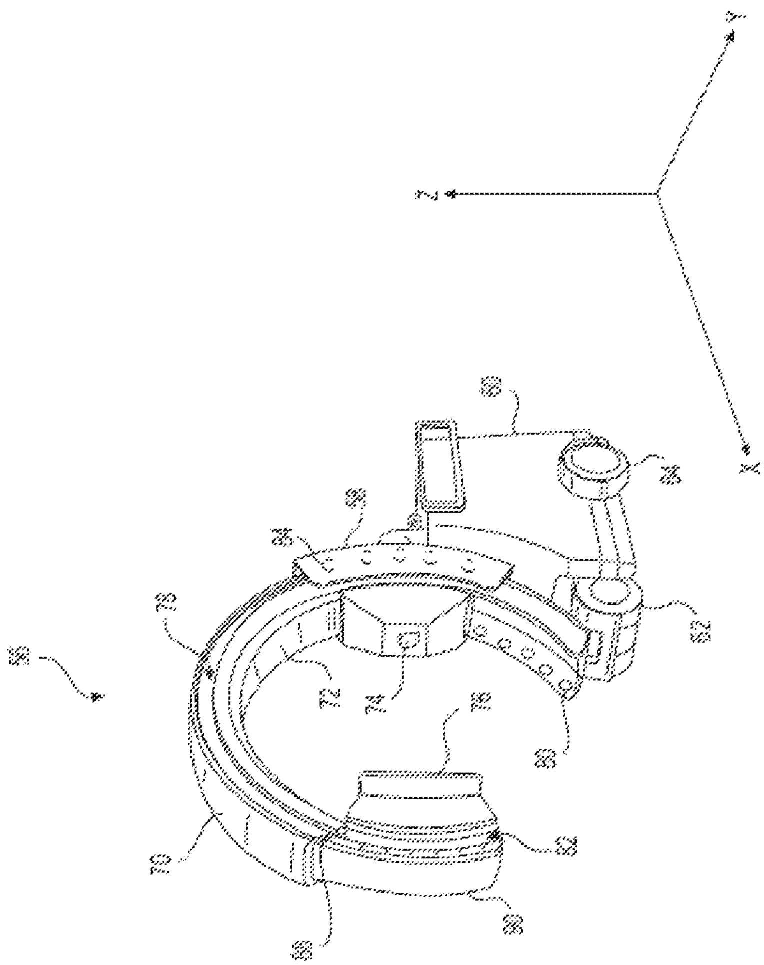

FIG. 3

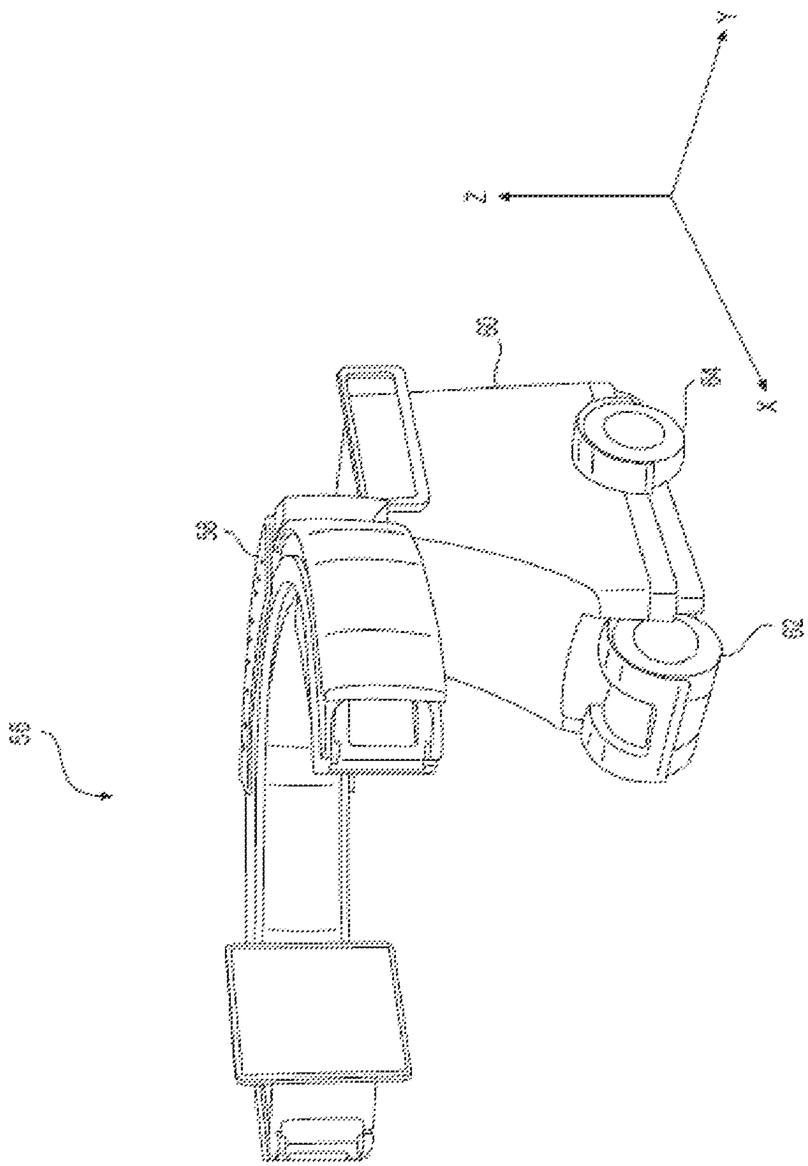

FIG. 4

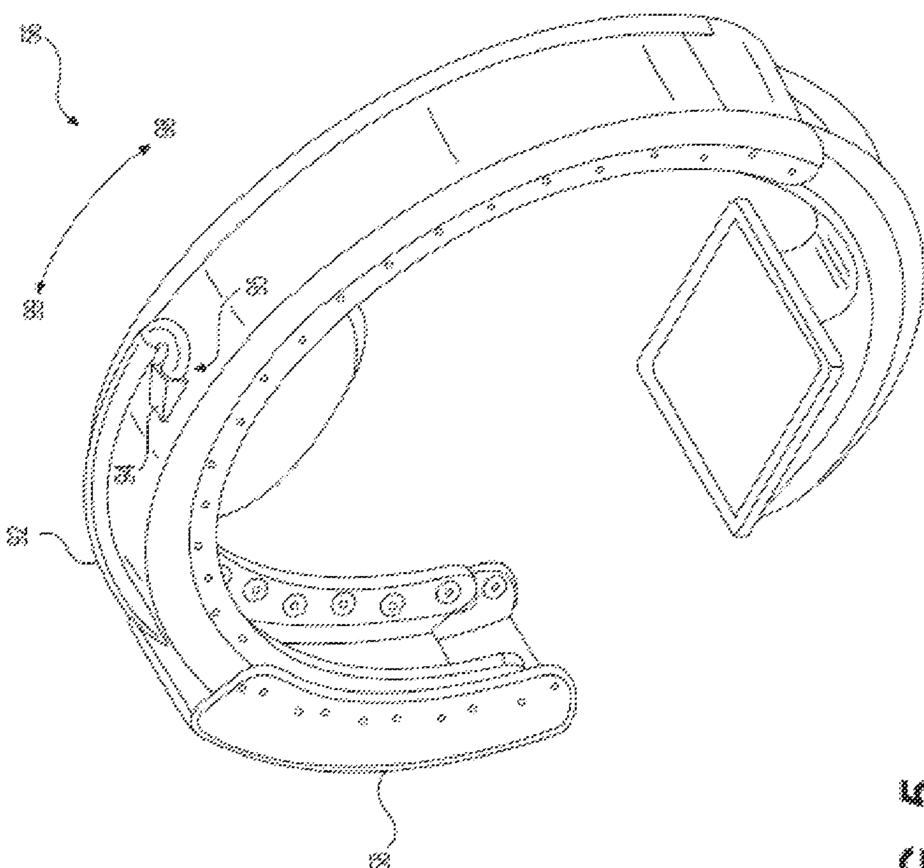

FIG. 5

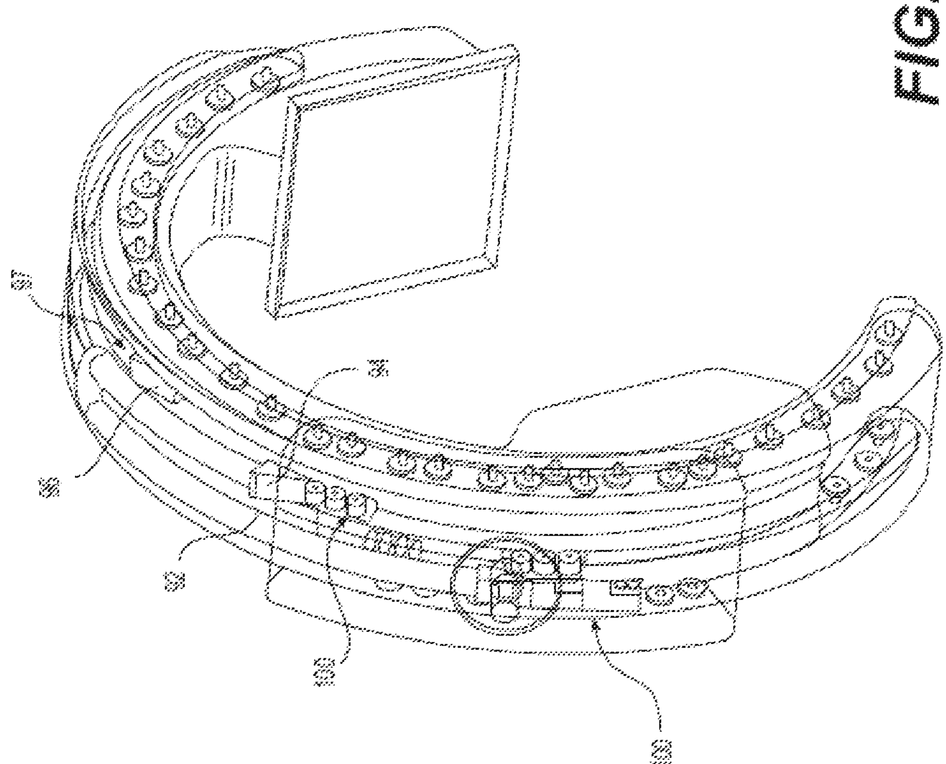

FIG. 6

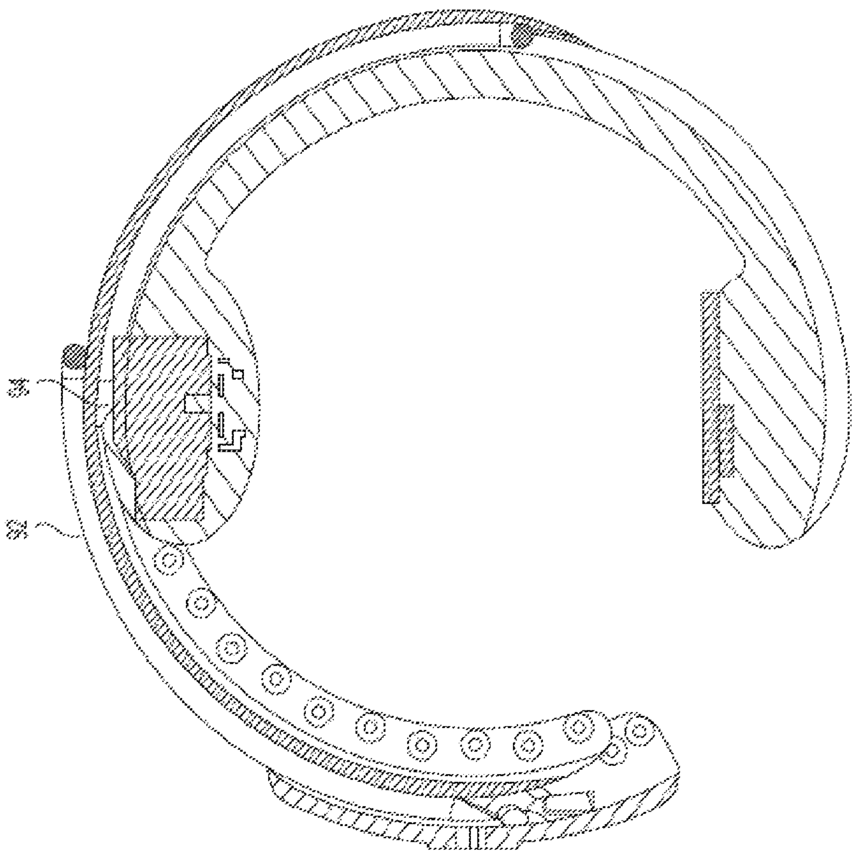

FIG. 7

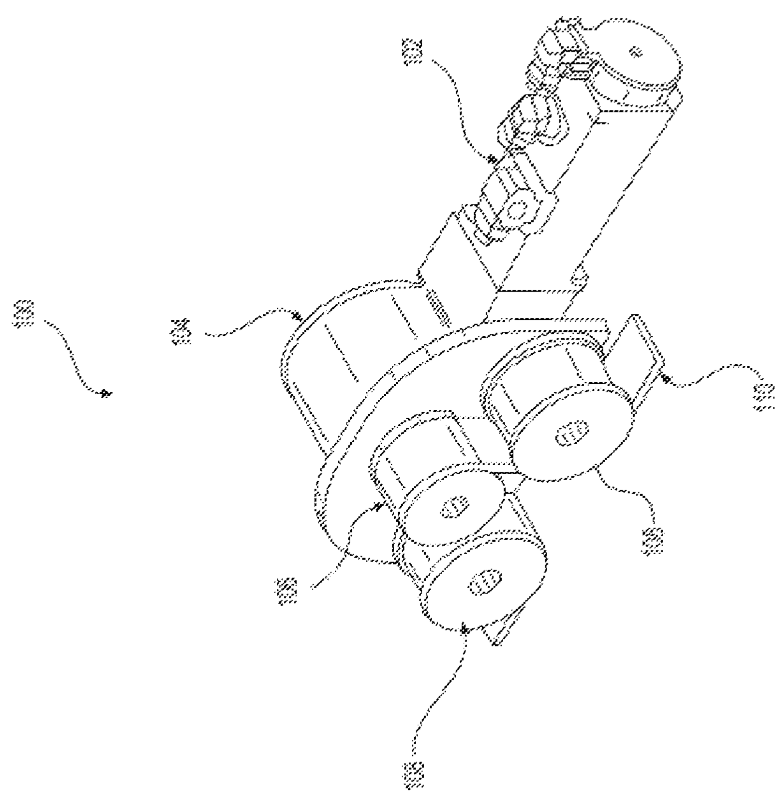

FIG. 8

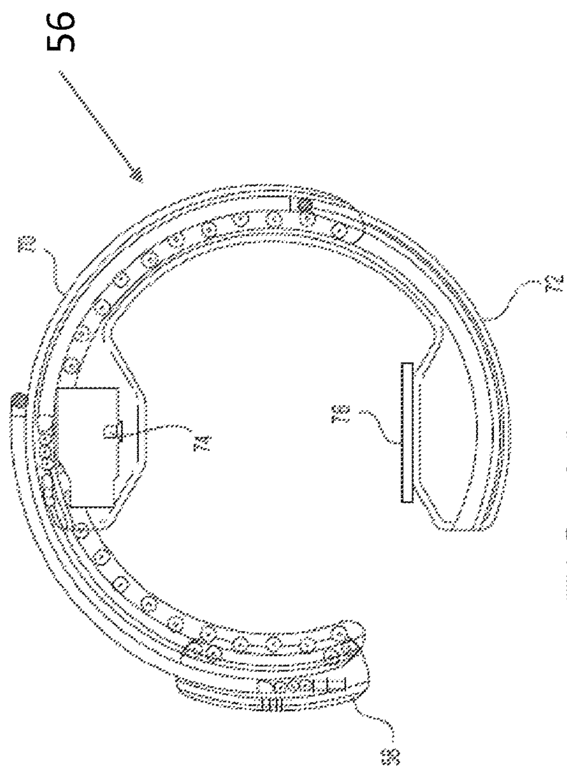

FIG. 9A

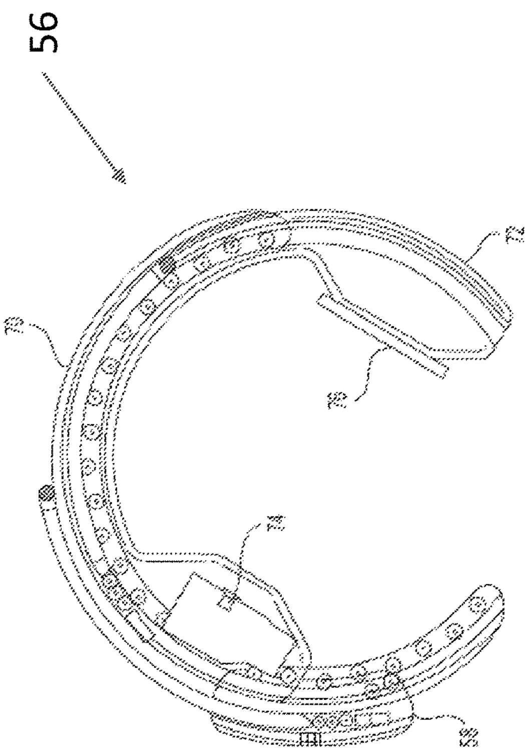

FIG. 9B

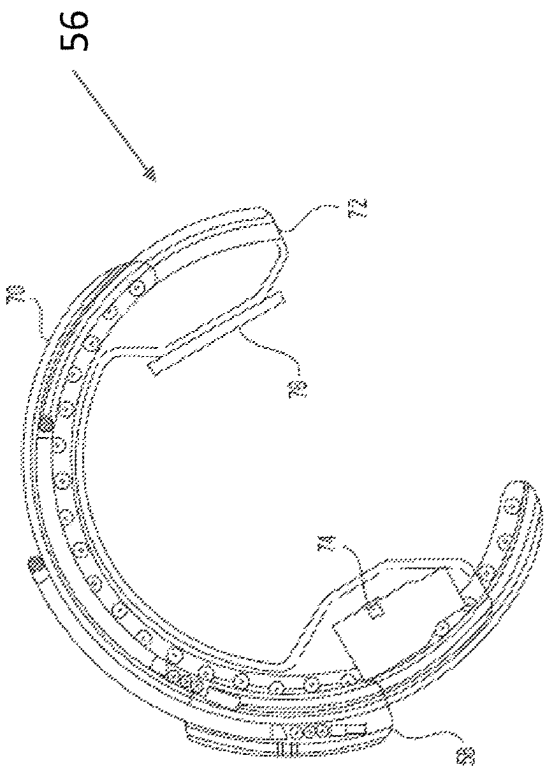

FIG. 9C

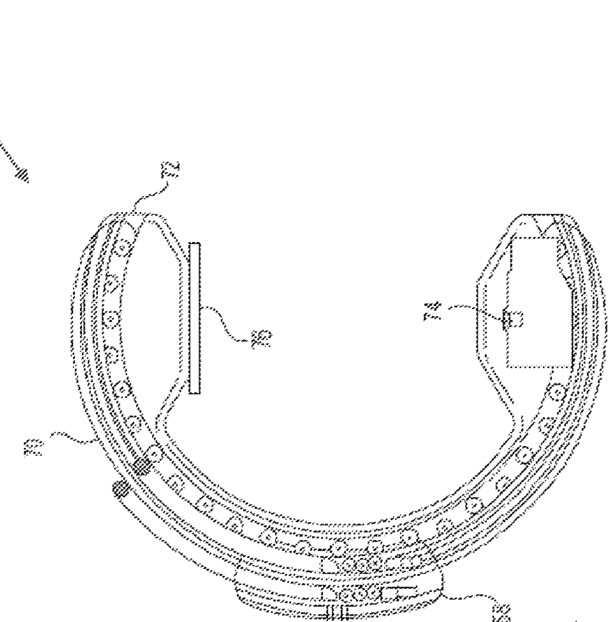

FIG. 9D

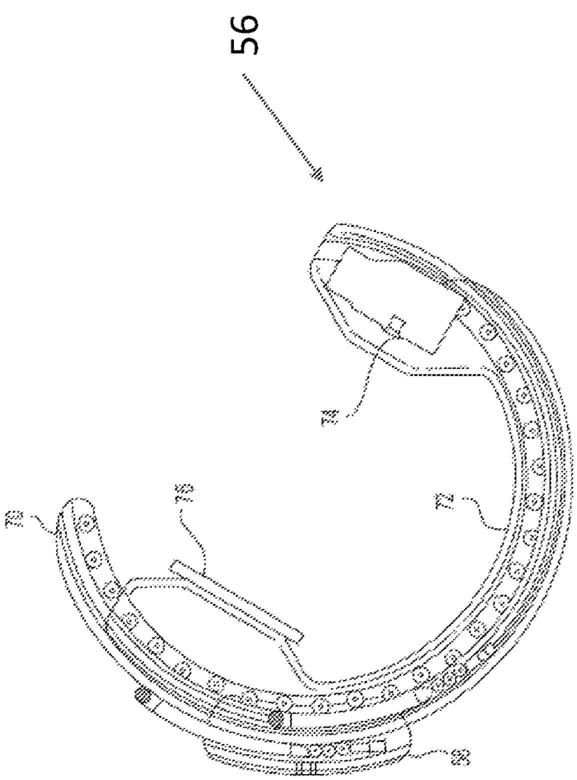

FIG. 9E

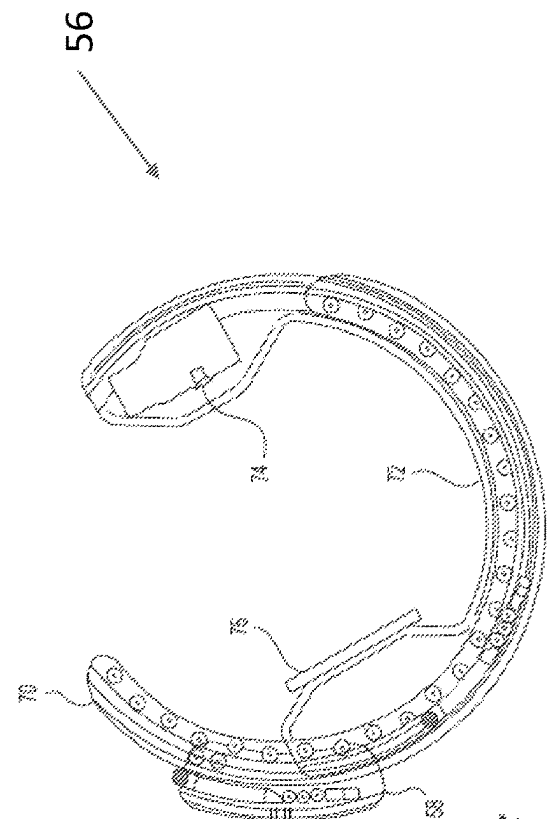

FIG. 9F

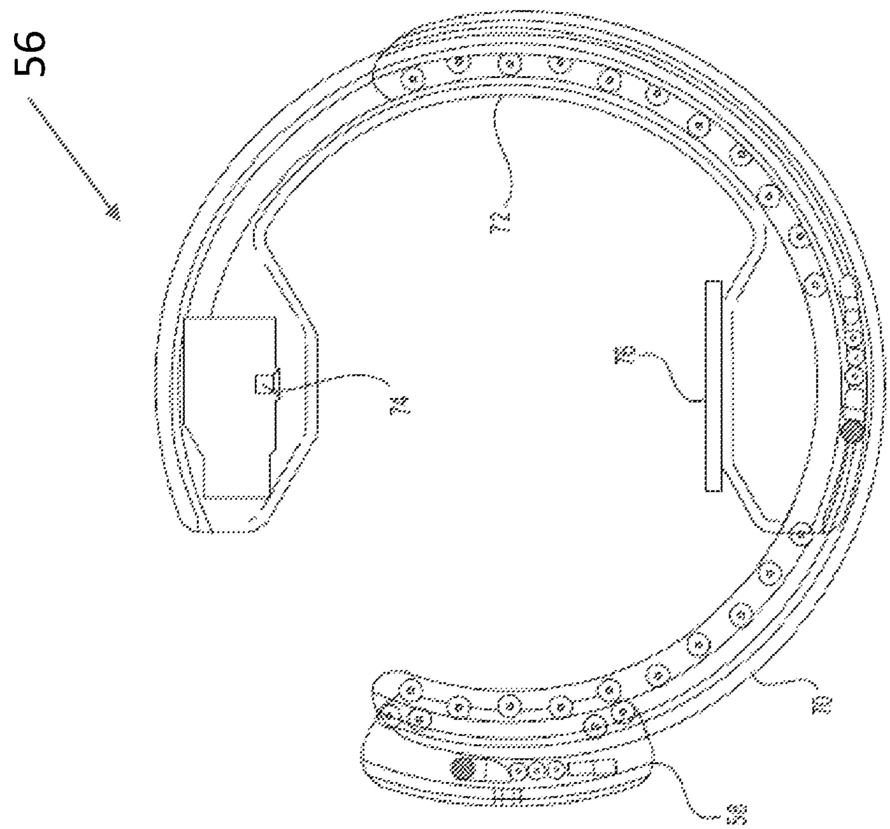

FIG. 9G

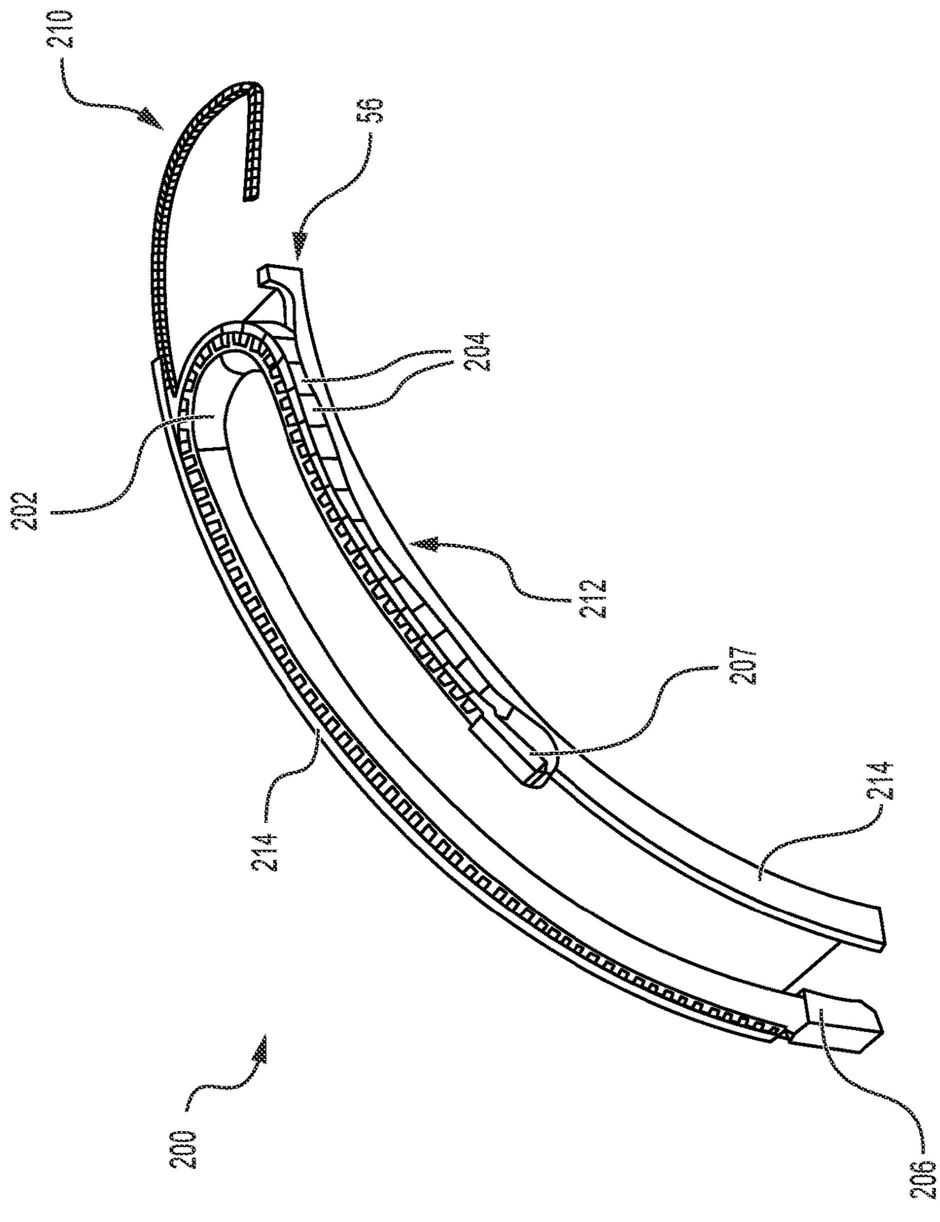

FIG. 10

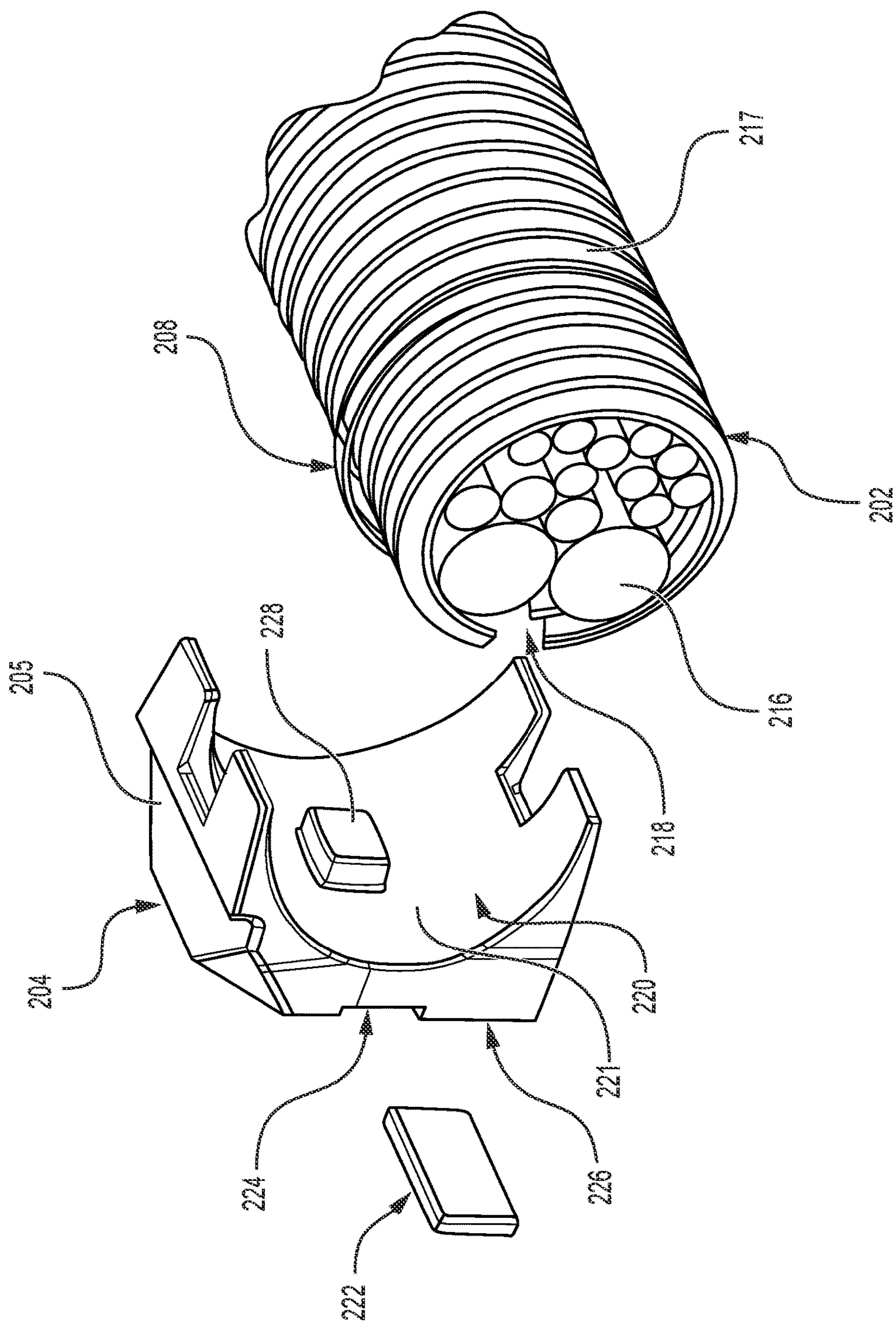

FIG. 11

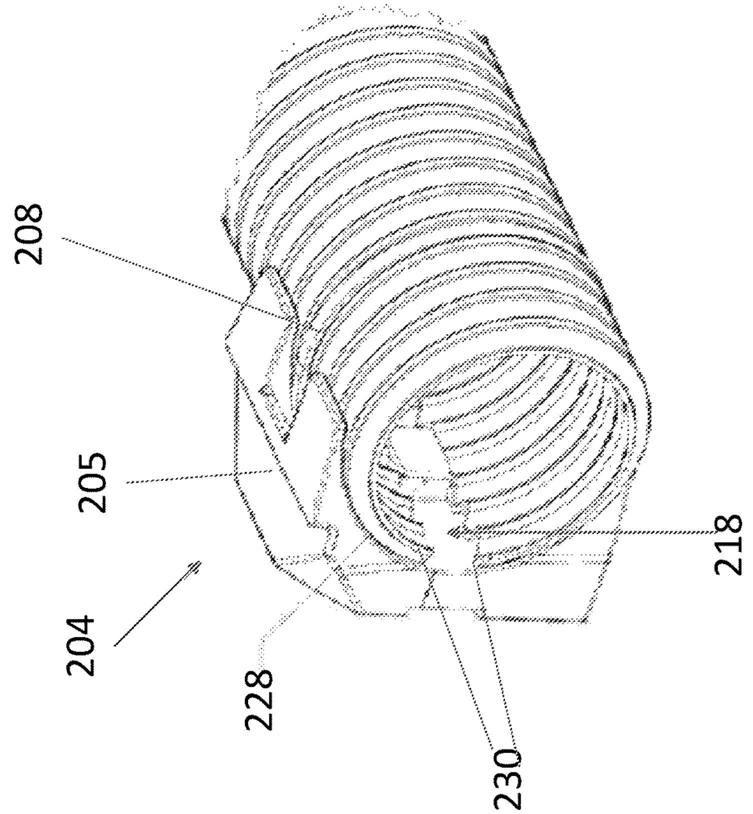

FIG. 12B

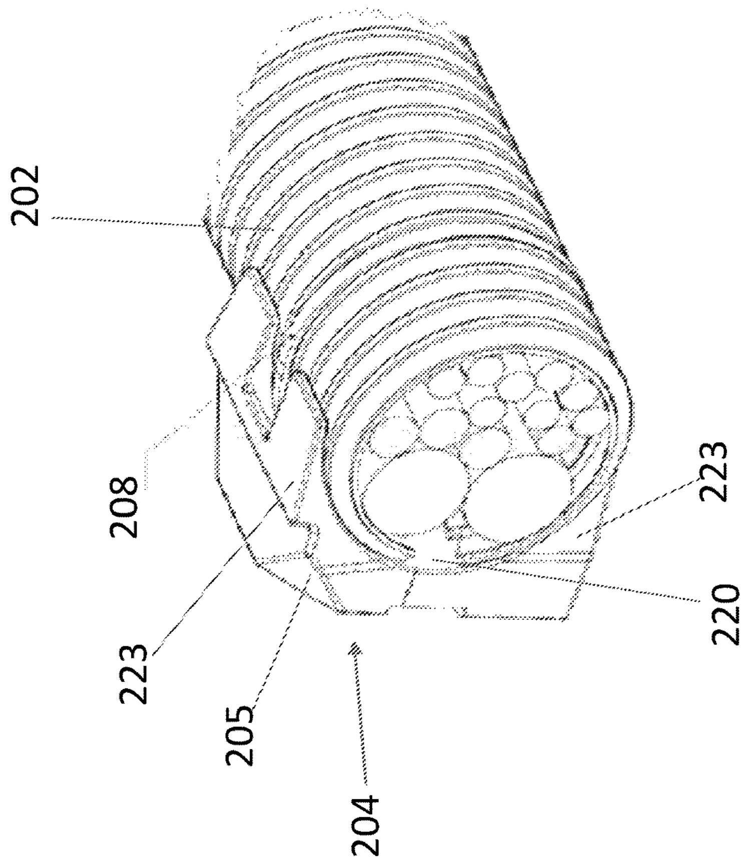

FIG. 12A

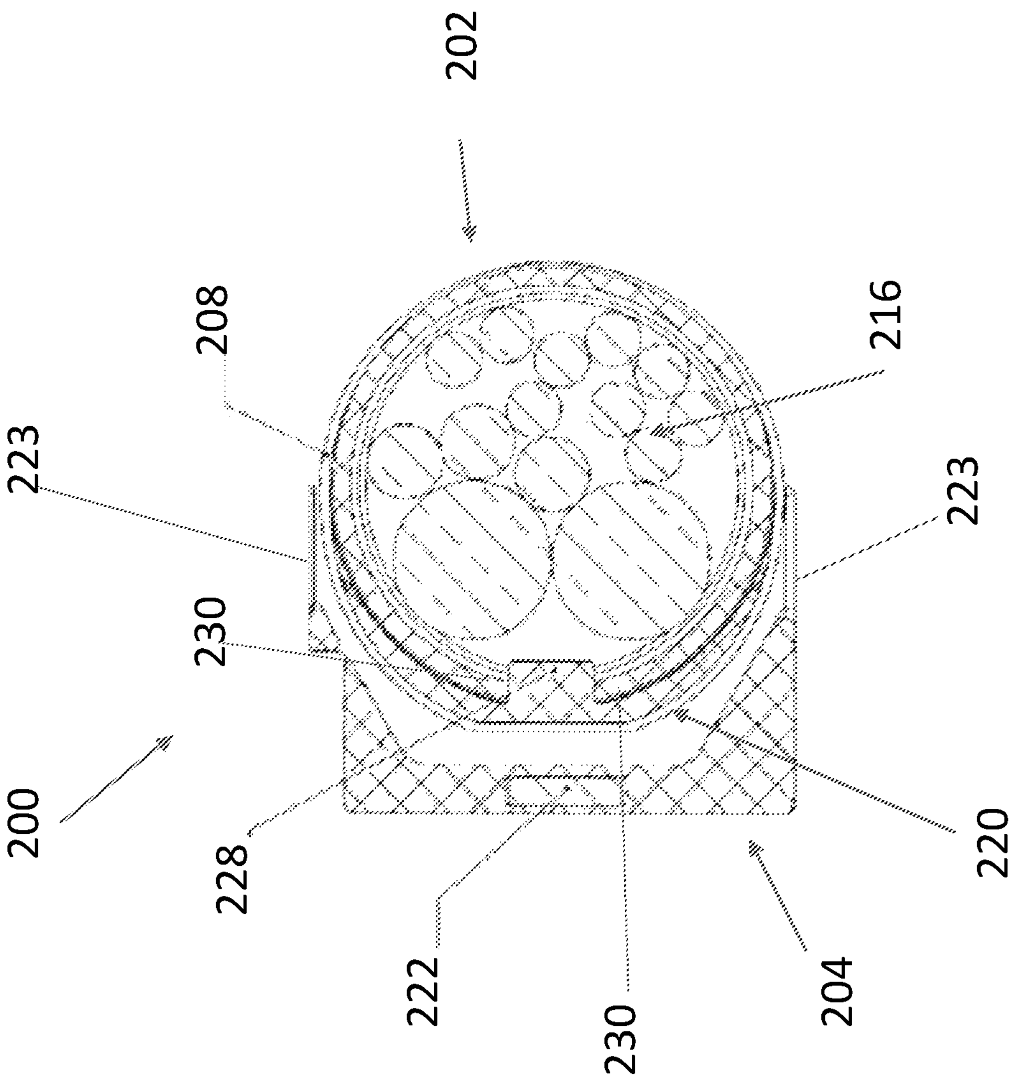

FIG. 13

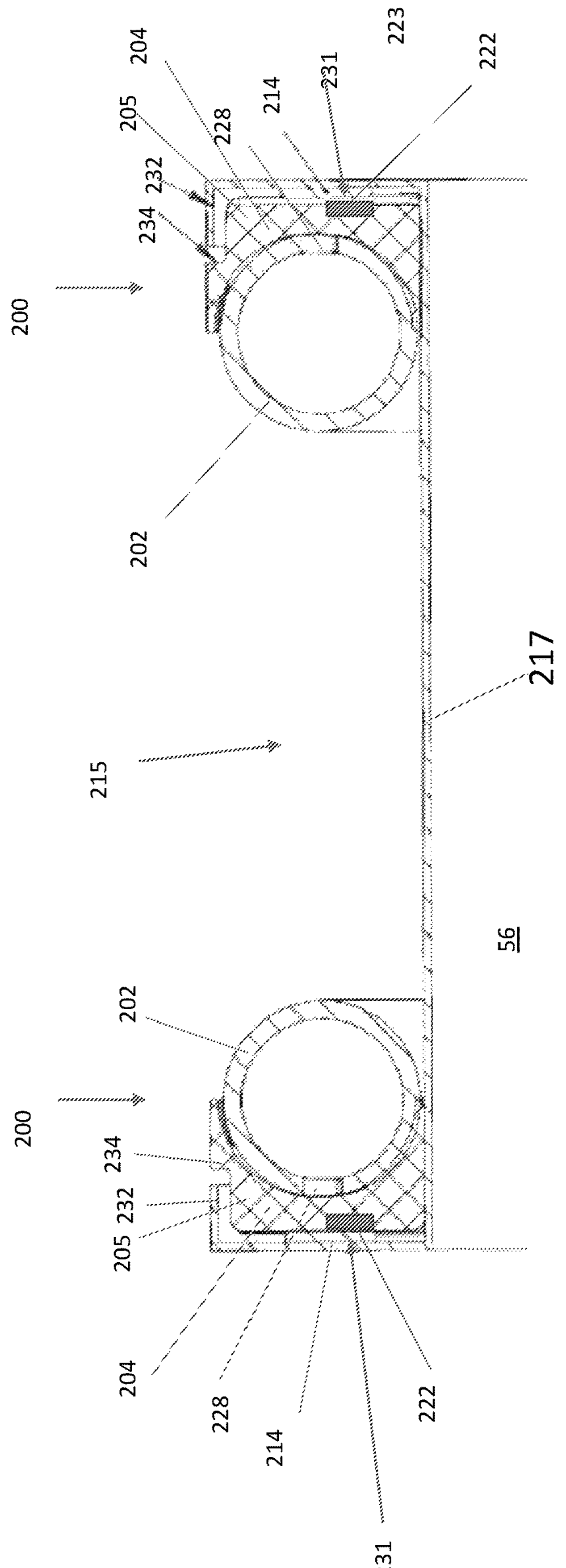
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FIG. 14

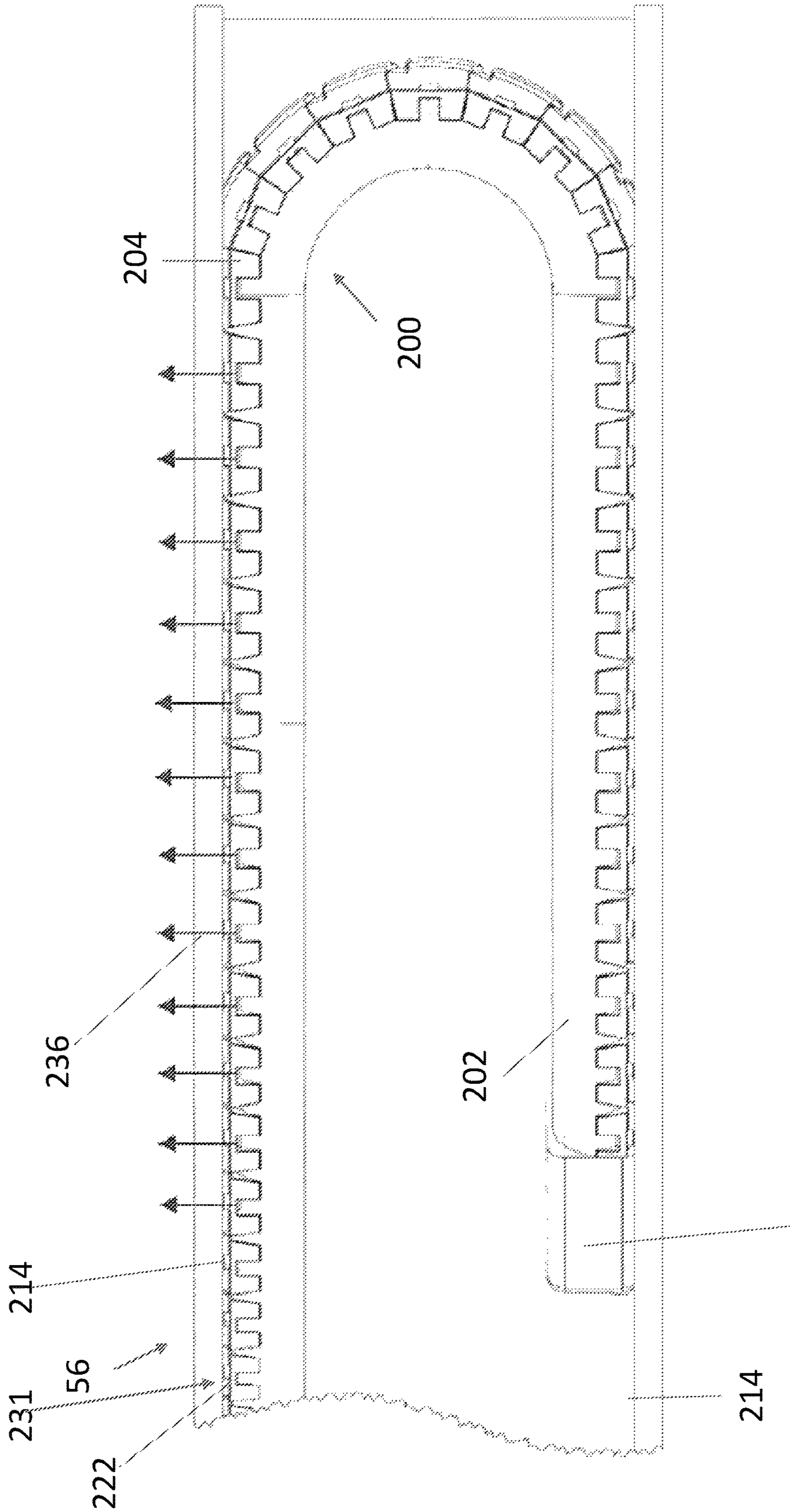

FIG. 15

COMPOUND CURVE CABLE CHAIN

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is a continuation of a U.S. patent application Ser. No. 17/009,504, filed on Sep. 1, 2020, which claims priority to provisional Patent Application Ser. No. 62/904,863 filed on Sep. 24, 2019, which is incorporated in its entirety herein.

BACKGROUND

Healthcare practices have shown a tremendous value of three-dimensional imaging such as computed tomography (CT) imaging. These imaging systems generally contain a fixed bore into which a patient enters from the head or foot. Other areas of care, including the operating room, intensive care departments and emergency departments, rely on two-dimensional imaging (fluoroscopy, ultrasound, 2-D mobile X-ray) as the primary means of diagnosis and therapeutic guidance. While mobile solutions for patient-centric 3-D imaging do exist, they are often limited by their freedom of movement to effectively position the system without moving the patient. Their limited freedom of movement has hindered an acceptance and use of mobile three-dimensional imaging systems.

Therefore, there is a need for mobile three-dimensional imaging systems for use in operating rooms, which can access the patients from any direction or height and produce high quality, three-dimensional images.

SUMMARY

In an exemplary embodiment, the present disclosure provides a cable chain comprising split tubing and a plurality of links extending discretely along a length of the split tubing. Each link comprises a housing including an outer surface and an inner surface. The outer surface comprises a magnet and the inner surface forms a recess in the link. The split tubing is disposed within the recesses of the links.

In another exemplary embodiment, the present disclosure provides a system comprising a cable chain that may include split tubing and a plurality of links extending discretely along a length of the split tubing. Each link may include a housing comprising an outer surface and an inner surface. The outer surface may comprise a magnet and the inner surface may form a recess in each of the links. The split tubing is disposed within the recesses of the links. A bundle of cables may be disposed within the split tubing. The system may also include a gantry. The cable chain may be movably disposed within the gantry.

In another exemplary embodiment, the present disclosure provides a system comprising first and second cable chains. Each cable chain may comprise split tubing and a plurality of links extending discretely along a length of the split tubing. Each link may comprise a housing including an outer surface and an inner surface. The outer surface may comprise a magnet and the inner surface may form a recess in the link. The split tubing is disposed within the recesses of the links. The system may also include a gantry comprising first and second sidewalls. The first cable chain is movably disposed along the first sidewall, and the second cable chain is movably disposed along the second sidewall.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional aspects, features, and advantages of the present disclosure will be apparent to one skilled in the art from the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

These drawings illustrate certain aspects of some of the embodiments of the present disclosure and should not be used to limit or define the disclosure.

FIG. **1** is a perspective rear view of an imaging system, in accordance with some embodiments of the present disclosure;

FIG. **2** is a schematic diagram of an imaging controller system, in accordance with some embodiments of the present disclosure.

FIG. **3** is a perspective front view of the imaging system, in accordance with some embodiments of the present disclosure;

FIG. **4** is a perspective view of the imaging system in which the gantry has been rotated about the X-axis by 90°, in accordance with some embodiments of the present disclosure;

FIGS. **5**-**7** illustrate perspective views of the gantry with a cabling arrangement, in accordance with some embodiments of the present disclosure;

FIG. **8** illustrates a motor assembly for telescopically controlling the C-arms of the gantry, in accordance with some embodiments of the present disclosure;

FIGS. **9**A-**9**G illustrate perspective views of a 360° rotation of the gantry in 60° increments, in accordance with some embodiments of the present disclosure;

FIG. **10** illustrates a perspective view of a cable chain, in accordance with embodiments of the present disclosure;

FIG. **11** illustrates a perspective view of the tubing and a link of the cable chain in accordance with embodiments of the present disclosure;

FIGS. **12**A and **12**B illustrate perspective views of the tubing assembled around the cable bundle, in accordance with some embodiments of the present disclosure;

FIG. **13** illustrates a cross-sectional front view depicting the cable chain, in accordance with some embodiments of the present disclosure;

FIG. **14** illustrates a cross-sectional front view depicting a nested geometry of multiple cable chains movably disposed along a gantry; and

FIG. **15** illustrates a perspective view of the cable chain movably disposed against the gantry, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the implementations illustrated in the drawings and specific language will be used to describe them. It will nevertheless be understood that no limitation of the scope of the disclosure may be intended. Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it may be fully contemplated that the features, components, and/or steps described with reference to one or more implementations may be combined with the features, components, and/or steps described with reference to other implementations of the present disclosure. For simplicity, in some instances the same reference numbers are used throughout the drawings to refer to the same or like parts.

Embodiments generally relate to a cable chain used to manage dynamic cable bundles routed through geographic information system (GIS) telescoping C-gantries ("gantries"). The cables may be routed internally through the gantries to facilitate 360° scanning of a patient. Routing cables externally may risk entangling the patient or a patient table during a 360° scan. Therefore, an internal cable management system is desired. More particularly, the cable chain may trace a compound curve rather than a planar bend (i.e., a path of the cable chain is curved when viewed from two planes). Further, the cable chain may operate in any orientation relative to gravity and may utilize a magnetic preload for attachment to the gantry. Additionally, the cable chain may have a low cross-sectional profile and may be able to turn along a tight bend radius to allow operation of the cable chain within a limited volume inside the gantry. The cable chain may ensure that the cable bundle follows a prescribed path and may also protect the cables through a full range of motion along the gantry. Absent a cable management system, the cable bundle may be prone to damage, sag under its own weight, intermittently causing interference issues, and/or jamming. These issues may be exacerbated depending on a tilt orientation of the gantry relative to gravity.

FIG. **1** is a schematic diagram showing an imaging system **10**, such as a computerized tomographic (CT) x-ray scanner, in accordance with embodiments of the present disclosure. The imaging system **10** may include a movable station **60** and a gantry **56**. The movable station may include a vertical shaft **59** and a gantry mount **58** which may be rotatably attached to the vertical shaft. The movable station **60** may include two front omni-directional wheels **62** and two rear omnidirectional wheels **64**, which together may provide movement of the movable station **60** in any direction in an X-Y plane. The omni-directional wheels **62**,**64** may be obtained, for example, from Active Robots Limited of Somerset, U.K. A pair of handles **13** mounted to the housing of the movable station **60** may allow a user to manually maneuver the station **60**. A motor **66** attached to the vertical shaft **59** may rotate the gantry mount **58** 360° about the X-axis, and a motor **67** may move the gantry mount **58** vertically along the z-axis under the control of the control module **51**. The gantry **56** may include a first C-arm **70** slidably coupled to the gantry mount **58** and a second C-arm **72** which may be slidably coupled to the first C-arm. In the embodiment shown, the first and second C-arms **70**,**72** are outer and inner C-arms, respectively. In the embodiment shown, the outer and inner C-arms **70**,**72** are circular in shape and rotate circumferentially about a central axis so as to allow imaging of a patient who is lying in bed **16** without the need to transfer the patient.

An imaging signal transmitter **74** such as an X-ray beam transmitter may be mounted to one side of the second C-arm **72** while an imaging sensor **74** such as an X-ray detector array is mounted to the other side of the second C-arm and faces the transmitter. In operation, the X-ray transmitter **74** transmits an X-ray beam which is received by the X-ray detector **76** after passing through a relevant portion of a patient (not shown). In one embodiment, the system **10** may be a multi-modality x-ray imaging system designed with surgery in mind. The three imaging modalities include fluoroscopy, 2D Radiography, and Cone-beam CT. Fluoroscopy is a medical imaging technique that shows a continuous X-ray image on a monitor, much like an X-ray movie. 2D Radiography is an imaging technique that uses X-rays to view the internal structure of a non-uniformly composed and opaque object such as the human body. CBCT (cone beam 3D imaging or cone beam computer tomography) also referred to as C-arm CT, is a medical imaging technique consisting of X-ray computed tomography where the X-rays are divergent, forming a cone. The movable station **60** may include an imaging controller system **40** which serves a dual function of (1) controlling the movement of the omni-directional wheels **62**,**64**, gantry mount **58** and the gantry **56** to position the imaging signal transmitter **74** in relation to the patient, and (2) controlling imaging functions for imaging the patient once the gantry **56** has been properly positioned.

FIG. **2** illustrates the imaging controller system **40** connected to a communication link **52**, in accordance with embodiments of the present disclosure. The imaging controller system **40** may be connected to the communication link **52** through an I/O interface **42** such as a USB (universal serial bus) interface, which receives information from and sends information over the communication link **52**. The imaging controller system **40** includes memory storage **44** such as RAM (random access memory), processor (CPU) **46**, program storage **48** such as ROM or EEPROM, and data storage **50** such as a hard disk, all commonly connected to each other through a bus **53**. The program storage **48** stores, among others, imaging control module **54** and motion control module **51**, each containing software to be executed by the processor **46**. The motion control module **51** executed by the processor **46** controls the wheels **62**,**64** of the movable station **60** and various motors in the gantry mount **58** and gantry **56** to position the station **60** near the patient and position the gantry in an appropriate position for imaging a relevant part of the patient. The imaging control module **54** executed by the processor **46** controls the imaging signal transmitter **74** and detector array **76** to image the patient body. In one embodiment, the imaging control module images different planar layers of the body and stores them in the memory **44**. In addition, the imaging control module **54** can process the stack of images stored in the memory **44** and generate a three-dimensional image. Alternatively, the stored images can be transmitted to a host system (not shown) for image processing.

The motion control module **51** and imaging control module **54** may include a user interface module that interacts with the user through the display devices **11***a* and **11***b* and input devices such as keyboard and buttons **12** and joystick **14**. Strain gauges **13** mounted to the handles **15** may be coupled to the I/O device **42** and conveniently provide movement of the movable station **12** in any direction (X, Y, Wag) while the user is holding the handles **15** by hand as will be discussed in more detail below. The user interface module assists the user in positioning the gantry **56**. Any of the software program modules in the program storage **48** and data from the data storage **50** can be transferred to the memory **44** as needed and is executed by the CPU **46**. The display device **11***a* is attached to the housing of the movable station **60** near the gantry mount **58** and display device **11***b* is coupled to the movable station through three rotatable display arms **16**, **18** and **20**. First display arm **16** is rotatably attached to the movable station **60**, second display arm **18** is rotatably attached to the first arm **16** and third display arm **20** is rotatably attached to the second display arm. The display devices **11***a*,**11***b* can have touch screens to also serve as input devices through the use of user interface modules in the modules **51** and **54** to provide maximum flexibility for the user.

Navigation markers **68** placed on the gantry mount **58** may be connected to the imaging controller system **40** through the link **52**. Under the control of the motion control module **51**, the markers **68** allow automatic or semi-automatic positioning of the gantry **56** in relation to the patient bed or OR (operating room) table via a navigation system (not shown). The markers **68** may be optical, electromagnetic or the like. Information can be provided by the navigation system to command the gantry **56** or system **10** to precise locations. One example may be that a surgeon holding a navigated probe at a desired orientation that tells the imaging system **10** to acquire a Fluoro or Radiographic image along that specified trajectory. Advantageously, this may remove the need for scout shots thus reducing x-ray exposure to the patient and OR staff. The navigation markers **68** on the gantry **56** may also allow for automatic registration of 2D or 3D images acquired by the system **10**. The markers **68** may also allow for precise repositioning of the system **10** in the event the patient has moved.

In the embodiment shown, the system **10** may provide a large range of motion in all 6-degrees of freedom ("DOF"). Under the control of the motion control module **51**, there are two main modes of motion: positioning of the movable station **60** and positioning of the gantry **56**. The movable station **60** positioning is accomplished via the four omni-directional wheels **62**,**64**. These wheels **62**,**64** allow the movable station **60** to be positioned in all three DOF about the horizontal plane (X, Y, Wag). "Wag" is a system **10** rotation about the vertical axis (Zaxis), "X" is a system forward and backward positioning along the X-axis, and "Y" is system **10** lateral motion along the Y-axis. Under the control of the control module **51**, the system **10** can be positioned in any combination of X, Y, and Wag (Wag about any arbitrary Z-axis due to use of omnidirectional wheels **62**,**64**) with unlimited range of motion. In particular, the omni-directional wheels **62**,**64** allow for positioning in tight spaces, narrow corridors, or for precisely traversing up and down the length of an OR table or patient bed.

The gantry **56** positioning may be accomplished about (Z, Tilt, Rotor). "Z" is gantry **56** vertical positioning, "Tilt" is rotation about the horizontal axis parallel to the X-axis as described above, and "Rotor" is rotation about the horizontal axis parallel to the Y-axis as described above. Together with the movable station **60** positioning and gantry **56** positioning, the system **10** provides a range of motion in all 6 DOF (X, Y, Wag, Z, Tilt and Rotor) to place the movable station **60** and the imaging transmitter **74** and sensor **76** precisely where they are needed. Advantageously, 3-D imaging can be performed regardless of whether the patient is standing up, sitting up or lying in bed and without having to move the patient. Precise positions of the system **10** can be stored in the storage memory **50** and recalled at any time by the motion control module **51**. This is not limited to gantry **56** positioning but also includes system **10** positioning due to the omni-directional wheels **62**,**64**.

FIG. **3** is a perspective front view of the imaging system of FIG. **1**, in accordance with embodiments of the present disclosure. Each of the gantry mount **58**, outer C-arm **70** and inner C-arm **72** respectively has a pair of side frames **86**, **88**,**90** that face each other. A plurality of uniformly spaced rollers **84** are mounted on the inner sides of the side frames **86** of the gantry mount **58**. The outer C-arm **70** has a pair of guide rails **78** on the outer sides of the side frames **88**. The rollers **84** are coupled to the guide rails **78**. As shown, the rollers **84** and the guide rails **78** are designed to allow the outer C-arm **78** to telescopically slide along the gantry mount **58** so as to allow at least 180 degree rotation of the C-arm about its central axis relative to the gantry mount. A plurality of uniformly spaced rollers **80** are mounted on the inner sides of the side frames **88** of the outer C-arm **70**. The inner C-arm **72** has a pair of guide rails **82** on the outer sides of the side frames **90**. The rollers **80** are coupled to the guide rails **82**. As shown, the rollers **80** and the guide rails **82** are designed to allow the inner C-arm **72** to telescopically slide along the outer C-arm **70** so as to allow at least 180 degree rotation of the C-arm about its central axis relative to the outer C-arm. Thus, the present invention as disclosed herein advantageously allows the gantry **56** to rotate about its central axis a full 360 degrees to provide the maximum flexibility in positioning the imaging system **10** with minimum disturbance of the patient. In another aspect of the present invention, a unique cabling arrangement is provided to make the imaging system **10** more compact and visually more appealing.

FIG. **4** is a perspective view of the imaging system **10**, in accordance with embodiments of the present disclosure. As illustrated, the gantry **56** has been rotated about the X-axis by 90°. Also illustrated are the movable station **60** and the gantry mount **58** for the gantry **56**. The movable station **60** may include two front omni-directional wheels **62** and two rear omnidirectional wheels **64**, which together may provide movement of the movable station **60** in any direction in an X-Y plane.

FIGS. **5**-**7** illustrate a cable carrier/harness **92**, in accordance with embodiments of the present disclosure. The cable carrier/harness **92** may contain electrical cables to carry signals between the imaging controller system **40** and various motors, X-ray transmitter **74**, imaging sensor **76** and various electronic circuits in the gantry **56**. A first cable router **94** is mounted to the outer surface of the outer C-arm **70** and a second cable router **96** is mounted to the outer surface of the inner C-arm **72**. Each cable router **94**,**96** has a through-hole **95**,**97** through which the cable carrier **92** passes. The cable carrier **92** extends from the gantry mount **56** over the outer surface of the first C-arm **70**, through the through-hole **95** of the first cable router **94** and over an outer surface of the second C-arm **72**. The cable carrier **92** overlying the first C-arm **70** extends in a first circumferential direction (clock-wise as shown) **98** and enters the first cable router **94** in a second circumferential direction (counter clock-wise as shown) **99** opposite to the first circumferential direction to create a 180 degree service loop over the outer surface of the first C-arm. From there, the cable carrier **92** extends in the first circumferential direction **98** and enters the second cable router in the second circumferential direction **99** to create another service loop over the outer surface of the second C-arm **72**. The particular locations of the first and second cable routers **94**,**96** combined with the service loops allow slack in the cable carrier **92** to provide the gantry **56** with full 360 degrees rotation without tangling or causing stress in the cable carrier. In the embodiment shown, the routers are mounted near the midpoint of the C-arms.

FIG. **8** illustrates one embodiment of a motor assembly **100** that could be used to telescopically rotate the outer C-arm **70** relative to the gantry mount **58** and inner C-arm **72** relative to the outer C-arm. Each motor assembly **100** includes a servo motor **102** with encoder feedback, gear box **104** to change the turning ratio, drive pulley **106**, idler pulleys **108** and belt **110** threaded between the drive pulley and the idler pulleys. One motor assembly **100** is mounted to the gantry mount to move the outer C-arm **70** relative to the gantry mount and another motor assembly is mounted to the outer C-arm **70** near the center of the arm to move the inner C-arm **70** relative to the outer C-arm.

FIGS. 9A-9G illustrate 360° rotation of the gantry **56** in the counterclockwise direction in 60° increments with FIG. **9**A representing a 0° position of the imaging sensor **76** and transmitter **74**. FIG. **9**B represents a 60° turn/position of the gantry **56**. FIGS. **9**C-**9**F illustrate further movement of the gantry **56**. For each 60° turn of the gantry **56**, the motor assemblies **100**, under the control of the motion control module **51**, turn the inner C-arm **72** by 30° counter-clock wise and also turn the outer C-arm **70** by 30° counter-clock wise for a combined 60° turn. FIG. **9**G represents a full 360° turn of the gantry **56**. As can be seen, the outer C-arm **70** and inner C-arm **72** have each moved 180° from the original 0° position of FIG. **9**A.

FIG. **10** illustrates a perspective view of a GIS cable chain **200** ("chain **200**") in accordance with some embodiments of the present disclosure. The chain **200** may include corrugated tubing **202** ("tubing **202**") and carrier links **204** ("links **204**") that may be coaxially aligned and positioned discretely along a length of the tubing **202**. The tubing **202** may be made of a flexible material such as plastic or rubber. The tubing **202** may contain or encompass an unbound or loose cable bundle (not shown). The links **204** may fit over or encompass at least a portion of the tubing **202**. The links **204** may extend longitudinally along the tubing **202** and directly abut one another in an end-to-end configuration, however, the links **204** are not rigidly hinged together, to allow the chain **200** to leverage flexibility of the tubing **202** and achieve a desirable or minimum bend radii. The chain **200** may include a first end clamp **206** that may be fixed to a stationary gantry mount (e.g., the gantry mount **58** shown on FIG. **1**). The chain **200** may also include a second end clamp **207** opposite to the first end clamp **206**. The second end clamp **207** may be attached to a moving gantry rolling interface or sidewall(s) **214** of the gantry **56** (also shown on FIG. **1** for example). The chain **200** may slide or otherwise move along the sidewalls **214** of the gantry **56**. Compound curves **210** and **212** represent different configurations or prescribed travel paths of the chain **200**, for example.

FIG. **11** illustrates a perspective view of the tubing **202** and a link **204** in accordance with some embodiments of the present disclosure. As illustrated, an outer surface of the tubing **202** may include ribs or corrugations **217** that extend along a circumference of the tubing **202**, in some examples. The tubing **202** may encompass a cable bundle **216**. In some embodiments, the tubing **202** may be split. For example, a slit or gap **218** may extend longitudinally along a wall of the tubing **202** (e.g., split tubing) to allow disposal of the cable bundle **216** within the tubing **202**, for example. That is, the gap **218** is in fluid communication with an interior of the tubing **202**. In some examples, a wire tie **208** may secure the tubing **202** against the bundle **216**, by tightening the wire tie **208**, for example. The wire tie **208** may be disposed between two corrugations **217**, for example. Each link **204** may each include a housing **205**. The housing **205** may include an inner surface or recess **220** that partially encompasses the tubing **202**. The recess **220** may be bound by a contoured or curved inner surface **221** to contact and receive the tubing **202**. The contours or curves of the tubing **202** and the recess **220** of the link **204** may complement each other to ensure a snug fit of the tubing **202** against the recess **220** of the link **204**.

In some embodiments, a permanent magnet **222** may be embedded into an exterior slot **224** of the housing **205** of each link **204** such that the magnet **222** is flush with an outer or exterior surface **226** of the link **204** to protect the magnet **222** from damage. Each link **204** may be magnetically attracted to ferrous material in the sidewall **214** (e.g., shown on FIG. **10**) which may ensure that the chain **200** (shown on FIG. **10**, for example) is preloaded into the sidewall **214**. The magnet **222** also assists in constraining a chain orientation of the chain **200** (e.g., shown on FIG. **10**) when each link **204** rolls from one sidewall **214** (e.g., shown on FIG. **10**) to another sidewall **214** of the gantry **56** (e.g., shown on FIG. **1**). In some embodiments, each magnet **222** may be sized to support a weight of the chain **200** against gravity when the chain **200** or a portion thereof, is in a horizontal orientation.

Each link **204** may be constrained rotationally around the tubing **202** with a tab **228** that protrudes from the recess **220** of each link **204**. For example, the tab **228** may extend into the gap **218** (e.g., split portion) of the tubing **202** to prevent rotation of the tubing **202** relative to the tab **228**. The tab **228** and the sidewall **214** (e.g., shown on FIG. **10**) may ensure that the tubing **202** resists undesired torsional motion or twisting. The tab **228** may also serve as an attachment point for the tubing **202**, in some examples.

FIG. **12**A illustrates a perspective view of the tubing **202** assembled around the cable bundle **216** and the link **204** assembled around the tubing **202**, in accordance with some embodiments of the present disclosure. The wire tie **208** may extend around or along a circumference of the tubing **202**. The tubing **202** may be disposed within the recess **220** of the link **204**. The link **204** may include extended portions **223** that may partially form the recess **220** and partially wrap around the tubing **202** to assist in securing the tubing **202**.

FIG. **12**B illustrates a perspective view of the tab **228** protruding from the recess **220** of the link **204**, in accordance with some embodiments of the present disclosure. The tab **228** is secured within the gap **218**. The gap **218** may be positioned between edges **230** of the tubing **202**. The edges **230** may extend along the length of the tubing **202** and squeeze the tab **228** to secure the tubing **202** to the link **204**, in some examples. The wire tie **208** may be tightened to compress or squeeze the edges **230** against the tab **228** to secure the tubing to the link **204**.

FIG. **13** illustrates a cross-sectional front view depicting the chain **200**, in accordance with some embodiments of the present disclosure. As illustrated, the link **204** only partially encompasses or covers the tubing **202**. This may cause the chain **200** to have a low-profile and allows greater flexibility for the chain **200** to turn within the gantry **56** (e.g., shown on FIG. **1**). The tubing **202** may be secured within the recess **220** of the link **204**. For example, the tubing **202** may be secured to the link **204** via a tightened wire tie **208** that compresses edges **230** of the tubing **202** against the tab **228**. In some embodiments, the magnet **222** may be adjacent to the tab **228**. Additionally, the cable bundle **216** may be disposed within the tubing **202**. In some embodiments, the tubing **202** may be disposed between the extended portions **223** of the link **204**.

FIG. **14** illustrates a cross-sectional front view depicting a nested geometry of multiple chains **200** movably disposed within or against the gantry **56**, in accordance with some embodiments of the present disclosure. A first chain **200** may be disposed at a first sidewall **214** of the gantry **56** and a second chain **200** may be disposed at a second sidewall **214** of the gantry **56**. The chains **200** may be separated by a void **215** that may extend longitudinally along the gantry **56**. In some embodiments, a surface **217** may extend between and adjacent to the sidewalls **214**. The surface **217** may continuously extend along a length of the gantry **56** and in between the chains **200**, as illustrated. The links **204** may be movably disposed against the sidewalls **214** such that the magnets **122** are adjacent to and/or in contact with ferrous material **231** disposed within the sidewalls **214**. In some embodiments, the links **204** may be attached to the sidewalls **214** via the magnets **222** and ferrous material **231**. Each of the sidewalls **214** may also include a flange **132** to mate with a step **234** of each housing **205** of the links **204** to assist in retaining the chains **200** against the sidewalls **214**, as illustrated. The links **204** may move along the length of the sidewalls **214** and therefore along a length of the gantry **56**. The tubing **202** may be attached to the links **204** via compression of the tabs **228** with the edges **230** of the tubing **202**, as previously noted. Accordingly, each of the chains **200** is configured to move independently along the length of the gantry **56**, in some examples.

FIG. **15** illustrates the chain **200** movably disposed within the gantry **56**, in accordance with embodiments of the present disclosure. As illustrated, the chain **200** may be in a horizontal position and the weight of the chain **200** is unsupported by the gantry **56**, however, discrete magnetic forces indicated by directional arrows **236**, support the weight of the chain **200** against the force of gravity. The directional arrows **236** indicate a direction that the chain **200** is pulled due to a magnetic attraction between the magnets **222** of the links **204** and the ferrous material **231** disposed within the sidewalls **214**. For example, the tubing **202** is attached to the links **204** to form the chain **200** that is movably attached the sidewalls **214** of the gantry **56**. The magnets **222** secure the chain **200** to the sidewalls **214** and in some embodiments, the chain **200** is pulled against the sidewalls **214** due to the magnetic forces. Additionally, the second end clamp **207** may be movably attached to the sidewalls **214** of the gantry **56**, as previously noted.

As described herein, some benefits of the various embodiments include: (1) a chain with a low profile and improved flexibility; (2) a path of the chain follows a compound curve; (3) the chain functions independent of orientation due to a magnetic preload; and (4) the split tubing allows for installation around a cable bundle rather than threading the cable bundle through a completely enclosed tube, and also allows cable bundles to be pre-terminated with connectors rather than terminated after assembly.

What is claimed is:

1. An imaging device having a cable chain assembly comprising:
- a c-arm having a first side wall and a second side wall, wherein the first and second side walls are opposing each other;
- a tubing adapted to house a plurality of cables; and
- a plurality of discrete carrier links extending along a length of the tubing, each link including a housing having an outer surface disposed to face a corresponding one of the side walls and a curved inner surface that circumferentially surrounds the tubing, but only partially circumferentially surrounds the tubing.

2. The imaging device of claim **1**, wherein the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing.

3. The imaging device of claim **1**, wherein each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the c-arm.

4. The imaging device of claim **3**, wherein the magnets of the links are sufficiently strong to provide a self-support of the tubing against gravity regardless of any angular orientation of the c-arm.

5. The imaging device of claim **1**, wherein the tubing is corrugated.

6. The imaging device of claim **1**, wherein each link includes a tie that wraps around the housing to couple the housing to the tubing.

7. The imaging device of claim **1**, wherein:
- the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing;
- each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the c-arm; and
- the tubing is corrugated.

8. The imaging device of claim **1**, wherein the tubing is a split tubing and the split of the split tubing faces inner surfaces of the links.

9. The imaging device of claim **8**, wherein each link includes a tab protruding from the inner surface and is received in the split of the housing.

10. The imaging device of claim **1**, wherein:
- the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing; and
- each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the c-arm such that the tubing is self-supported against gravity regardless of any angular orientation of the c-arm.

11. An imaging device having a cable chain assembly comprising:
- an outer c-arm having a first side wall and a second side wall, wherein the first and second side walls oppose one another;
- an inner c-arm having first second opposing side walls, and slidably coupled to the outer c-arm, the outer c-arm and inner c-arm together configured to provide a 360 degree rotation;
- wherein each c-arm includes:
  - a tubing adapted to house a plurality of cables; and
  - a plurality of discrete carrier links extending along a length of the tubing, each link including a housing having an outer surface disposed to face a corresponding one of the side walls and a curved inner surface that circumferentially surrounds the tubing, but only partially circumferentially surrounds the tubing.

12. The imaging device of claim **11**, wherein the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing.

13. The imaging device of claim **11**, wherein each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the corresponding c-arm.

14. The imaging device of claim **13**, wherein the magnets of the links are sufficiently strong to provide a self-support of the tubing against gravity regardless of any angular orientation of the corresponding c-arm.

15. The imaging device of claim **11**, wherein the tubing for each c-arm is corrugated.

16. The imaging device of claim **11**, wherein each link includes a tie that wraps around the housing to couple the housing to the tubing.

17. The imaging device of claim **11**, wherein:
- the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing;

each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the corresponding c-arm; and

the tubing for each c-arm is corrugated.

18. The imaging device of claim **11**, wherein the tubing for each c-arm is a split tubing and the split of the split tubing faces inner surfaces of the links.

19. The imaging device of claim **18**, wherein each link includes a tab protruding from the inner surface and is received in the split of the housing.

20. The imaging device of claim **11**, wherein:

the links are disposed adjacent to each other without any hinged connection to each other to provide a tighter bending radius of the tubing; and

each link includes a magnet housed in the housing for attraction towards the corresponding sidewall of the corresponding c-arm such that the tubing is self-supported against gravity regardless of any angular orientation of the corresponding c-arm.

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